

CASE STUDY OF SERVICE RETAIL STORE

A Mid-Term Report for the BDM Capstone Project

Submitted by

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1. Executive Summary

This project is based on Service Retail Store, a small retail outlet that stocks a mix of everyday essentials (personal care and cleaning products), packaged food and snacks, and stationery items — catering to students, hostel residents, local residents, and staff. The store has built a loyal walk-in customer base, but faces three recurring operational issues, inventory imbalance across categories, margin leakage despite steady sales, and underused cross-category selling potential. This analysis uses data-driven methods to quantify each of these three issues and recommend corrective action.

Primary data was collected over an 85-day period (02 May – 25 July 2026) across 71 products spanning four categories — Food, Cleaning, Personal Care, and Stationery (grouped in parts as Essentials = Cleaning + Personal Care), Food, and Stationery . The dataset spans three linked sheets: Sales , Purchase and Inventory (141 physical stock-count entries collected on a rolling cycle-count basis).

From this data, key insights emerged. Essentials generated the highest total revenue (₹47790) at a 21.99% margin; Stationery, despite the lowest revenue (₹8976), carried the highest margin (31.74%) — driven by a mean transaction-level margin of 34.94%, well above the 21.7–22.4% seen in the other three categories; Food sat at the lowest average margin among the four individual categories (21.68%) and also the highest wastage, while being the category most prone to running out of stock — Food accounted for 12 of the 23 stockout instances recorded, against an average stock cover of just 12.3 days, compared to 48–54 days of cover sitting in Cleaning, Personal Care, and Stationery.

Overall, the store generated ₹91,675 in revenue against ₹71,465 in cost of goods sold, a gross profit of ₹20,210 and net margin (19.2% margin) over the period assuming the fixed cost of electricity and rent combined been Rs 2650 according to owners statement.

The project expected outcomes is rebalance restocking frequency so Food gets replenished more often while Cleaning, Personal Care, and Stationery purchase quantities are trimmed; investigate why Food's margin lags despite premium perishable pricing (spoilage is a likely driver); and test bundling between Food and Personal Care items, which co-occur in the same customer's purchases most often.

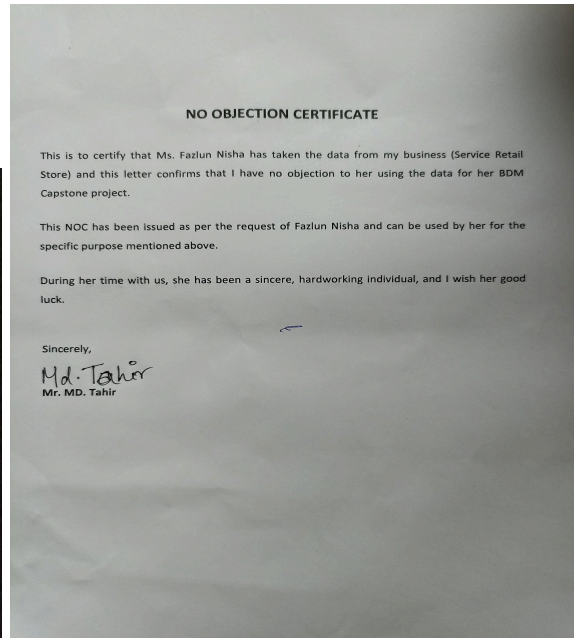
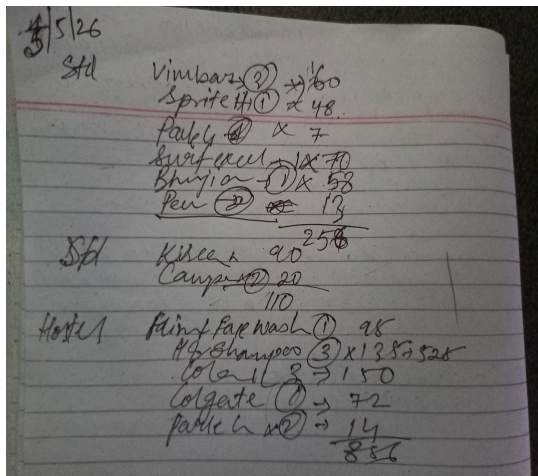
2. Proof of Originality of the Data

Business Name: Service Retail Store

Business Owner: Md. Tahir

Link: [proof of Originality](#)

Business Address: 21, Dedar Buksh Lane, Kolkata-700016



3. Metadata and Descriptive Statistics

3.1. Metadata

Metadata refers to “data about the data” — structured information about each variable collected, its measurement, and its relevance to the business problem. Data was collected over an 85-day period (02 May – 27 July 2026) for 71 SKUs across four categories: Food, Cleaning, Personal Care, and Stationery. The dataset consists of three linked sheets: Sales, Purchase, and Inventory.

Link of Dataset

Sales sheet :

- Date (DD-MM-YYYY) — day of the transaction.
- Customer_Type — Student, Hostel Resident, Local Resident, or Staff; was asked by me to shop owner to keep record for cross selling analysis.
- Product_Name / Category — item identity and category grouping.
- Quantity_Sold — units sold in that transaction line.
- Cost_Price_per_Unit / Selling_Price_per_Unit / Profit_per_Unit .
- Total_Cost / Total_Revenue / Total_Profit / Profit_Margin_Pct .

Purchase sheet :

- Purchase_ID, Date, Product_Name, Category, Supplier.
- Quantity_Purchased, Cost_Price_per_Unit, Total_Purchase_Cost.
- Expiry_Date — used to flag perishability risk, mainly for Food and some Personal Care items.

Inventory sheet :

- Date, Product_Name, Category.
- Opening_Stock_Units, Units_Purchased, Units_Sold, Units_Wasted_Damaged — the period’s flow.
- Closing_Stock_System_Units — the expected stock level, computed as Opening + Purchased – Sold – Wasted.
- Physical_Stock_Count_Units — the actual counted stock; small gaps versus the system figure reflect normal manual counting error and shrinkage.

Together, these three sheets allow revenue and cost to be traced back to physical stock movement, which is what makes it possible to separate high revenue products from high profit, efficiently-stocked products.

3.2. Descriptive Statistics

Descriptive statistics — count, mean, median, mode, standard deviation, minimum, and maximum — were computed at the transaction level for Quantity_Sold, Total_Revenue, and Profit_Margin_Pct, broken down by each of the four actual product categories (Food, Cleaning, Personal Care, Stationery), rather than pooled across the whole store. A single store-wide average would blur exactly the category-level differences this project needs to detect, so every statistic below is reported per category.

Table 3.2.1 — Quantity_Sold by Category

Category	Count	Mean	Median	Mode	Std_Dev	Min	Max
Food	336	2.29	2	1	1.54	1	9
Cleaning	146	1.82	1	1	1.24	1	7
Personal Care	218	1.72	1	1	1.32	1	7
Stationery	224	2.00	2	1	1.42	1	10

Table 3.2.1 — Distribution of units sold per transaction line, by category

Food shows the highest mean units per transaction (2.29), consistent with its role as the fastest-moving, highest-frequency category. Personal Care has the lowest mean (1.72) — customers typically buy one unit at a time. All four categories are right-skewed (mode = median = 1–2, but the mean is pulled up by occasional bulk purchases), a pattern that recurs across every table in this section.

Table 3.2.2 — Total_Revenue by Category (₹, transaction-level)

Category	Count	Mean	Median	Mode	Std_Dev	Min	Max
Food	336	103.90	65	48	101.997	7	700
Cleaning	146	142.88	95	140	125.861	20	570
Personal Care	218	123.53	87.5	65	98.408	32	665
Stationery	224	40.07	25	8	56.547	4	500

Table 3.2.2 — Distribution of per-transaction revenue, by category

Cleaning generates the highest average revenue per transaction (₹142.88), reflecting higher unit prices for detergents and cleaning agents, followed by Personal Care (₹123.53). Stationery has by far the lowest mean transaction value (₹40.07) — low-cost items like pens and pencils — but also the largest gap between median (₹25) and max (₹500), signalling occasional large bulk purchases.

Table 3.2.3 — Profit_Margin_Pct by Category (transaction-level)

Category	Count	Mean	Median	Mode	Std_Dev	Min	Max
Food	336	21.68%	21.00%	20%	4.13	15%	30%
Cleaning	146	22.32%	21.00%	21%	2.56	19%	29%
Personal Care	218	22.39%	22.00%	22%	1.34	20%	25%
Stationery	224	34.94%	33.00%	33%	5.39	25%	50%

)

Table 3.2.3 — Distribution of profit margin percentage, by category

This is the table most directly relevant to Problem Statement 2. Food, Cleaning, and Personal Care cluster tightly around a 22.4% mean margin with low standard deviation (1.3–4.1) — stable, predictable pricing. Stationery stands apart, both in level (34.94% mean margin, roughly 13 percentage points higher than the other three) and in spread (standard deviation 5.39, the widest of the four) — meaning Stationery pricing is both more profitable per rupee of sales and more variable than the rest of the store. This single finding directly informs the pricing recommendations.

4. Detailed Explanation of Analysis Process / Method

4.1. Data Cleaning and Preprocessing

Dates across all three sheets were parsed and standardised to a common format for chronological analysis. Product names and category labels were checked for consistency to avoid duplicate groupings during aggregation. Quantities were confirmed to be in whole units and prices in Indian Rupees per unit. Purchase and Inventory purchase figures were checked for internal consistency. To enable segment wise analysis of customer purchasing behaviour, I request the shopkeeper to record a customer category against each transaction this way the sales pattern against the customer segment is analysed even though the transaction id is not available.

4.2. Analysis Process, Tools, and Justification

Three complementary methods were used to address the three problem statements, chosen specifically for fit with a small retailer managing diverse categories with different demand cycles:

Problem Statement 1 (Inventory): ABC-style classification combined with reorder-point or days-of-stock-cover calculation. Days of Stock Cover = Physical Stock Count ÷ Average Daily Sales Rate, computed per product from its most recent physical count, then averaged by category. Stock Value Tied Up = Physical Stock Count × Unit Cost Price, summed by category,

to quantify working capital locked in slow-moving stock. This approach was chosen for its simplicity and its direct fit with a small store that cannot support complex forecasting models.

- Problem Statement 2 (Margins): Category-wise and product-wise descriptive statistics on Profit_Margin_Pct (Section 3.2, Table 3.2.3), combined with wastage tracking from the Inventory register, to isolate whether margin loss stems from pricing, procurement cost, or wastage — linking the margin analysis directly to the inventory findings rather than treating margin decline in isolation.
- Problem Statement 3 (Cross-selling): A customer-segment category-affinity analysis (percentage of each customer segment's spend by category) combined with a same-day, same customer-type co-purchase proxy, used because no transaction/basket ID was available for small retail store.

Key formulas used across the analysis:

- Revenue = Selling Price $\times$ Quantity Sold, aggregated by category and by transaction.
- Gross Profit = Total Revenue – Total Cost; Profit Margin (%) = (Profit $\div$ Revenue) $\times$ 100.
- Days of Stock Cover = Physical Stock Count $\div$ Average Daily Sales Rate (per product, then averaged by category).

Microsoft Excel was used to organize, clean, and structure the raw data:

SUMIFS/AVERAGEIFS/MAXIFS formulas, PivotTable-style category summaries, and conditional formatting were used to filter key trends, group data by category and customer type, and generate the charts used throughout this report. Where a metric needed to be computed within a category subset — median, mode, and standard deviation by category — array formulas were used in place of PivotTables to keep every figure formula-linked and auditable back to the raw Sales, Purchase, and Inventory sheets.

5. Results and Findings

5.1. Revenue and Profitability Patterns

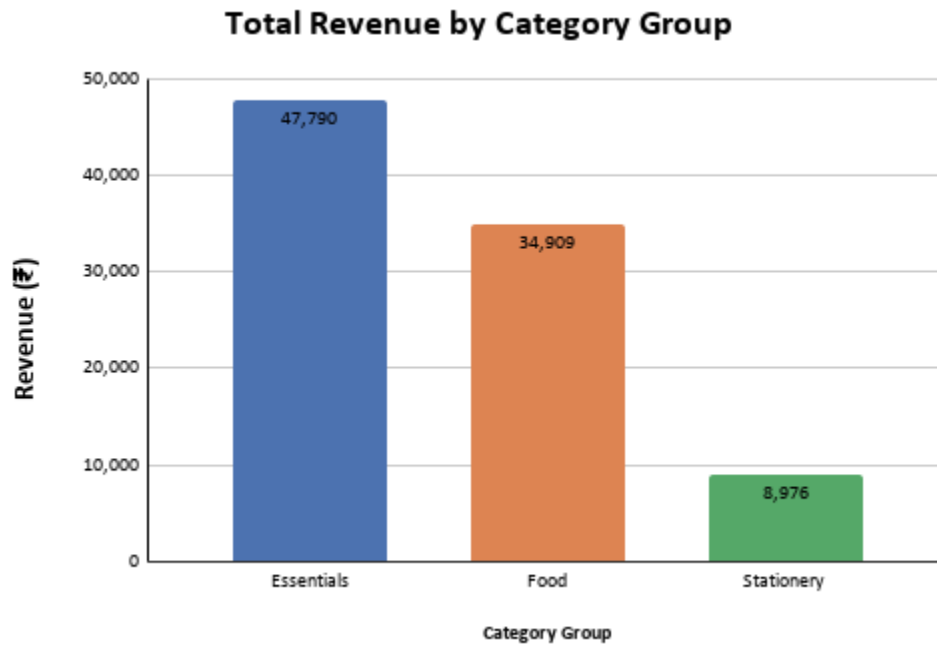


Figure 1 — Total revenue by category group over the 85-day period

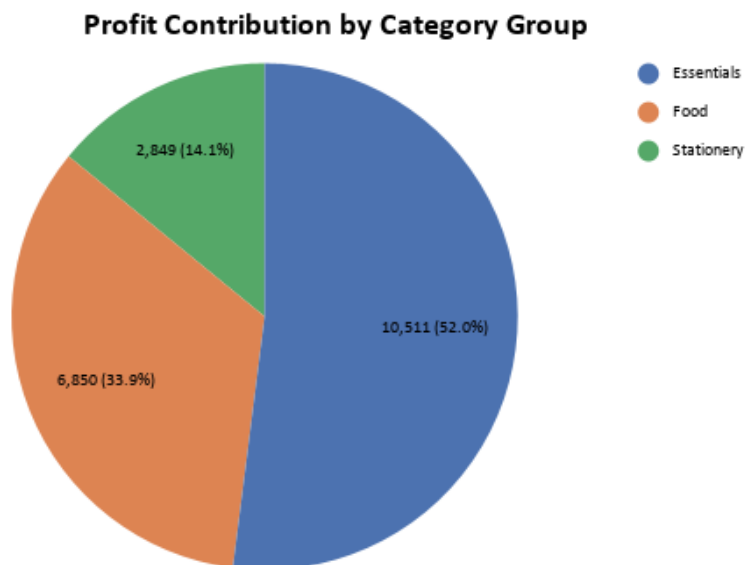


Figure 2 — Profit contribution by category group

Note that the category margins above are revenue-weighted (total category profit ÷ total category revenue) rather than the simple transaction-level average shown in Table 3.2.3 — this is why, for example, Food's margin here (19.62%) differs from its transaction-level mean (21.68%): larger-value Food transactions carry a slightly lower margin than smaller ones, pulling the revenue-weighted figure down. Essentials (Cleaning + Personal Care) is the store's revenue backbone, contributing ₹47,790 (52.1% of total revenue) at a 21.99% margin. Food contributes ₹34,909 (38.1%) but at a lower margin, 19.62%. Stationery is the smallest revenue contributor (₹8,976, 9.8%) but the most profitable per rupee of sales, at 31.74% margin — consistent with the transaction-level margin gap already identified in Table 3.2.3. Overall gross profit across the store was ₹20,210 on ₹91,675 of revenue (19.2% net margin) by assuming Fixed operating costs (rent, electricity,) is Rs 2650 over three months are based on owner's estimate, not captured in the sales/inventory data.

5.2. Inventory Findings — Problem Statement 1

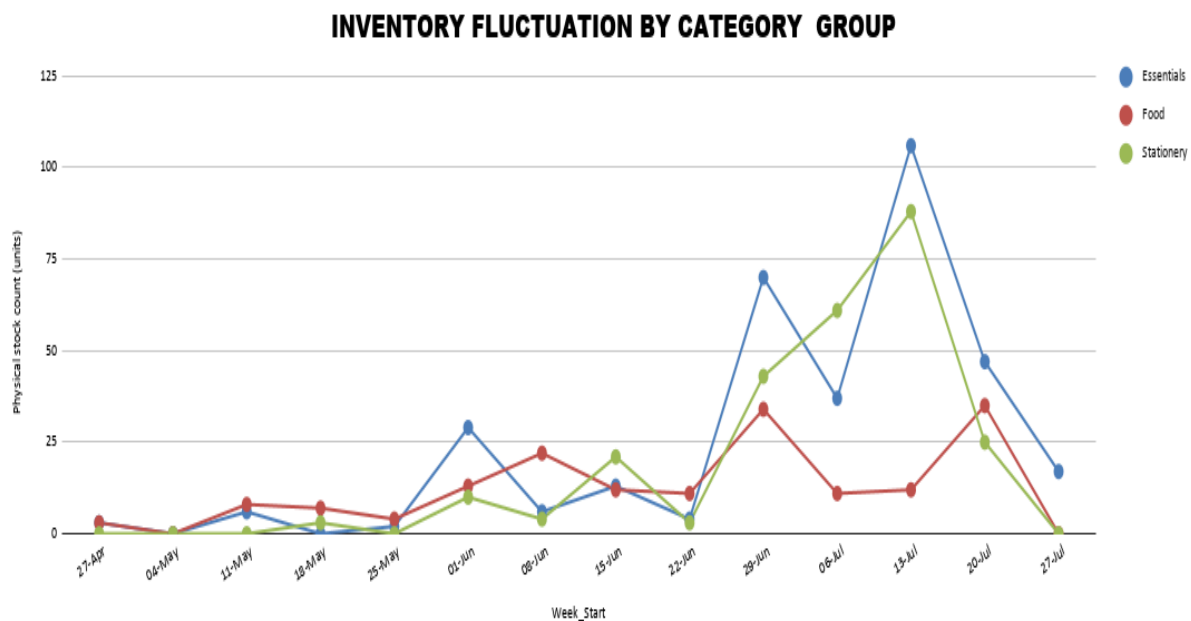


Figure 3 — Weekly stock-take totals by category group. Spikes reflect weeks when slow-moving Cleaning/Personal Care/Stationery items received their (infrequent) physical count, not real stock surges.

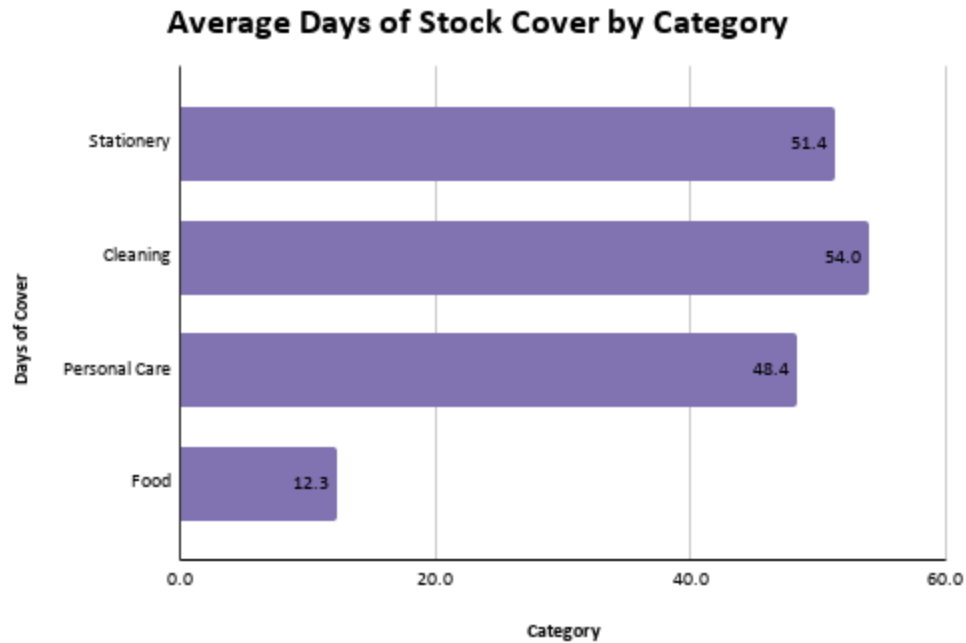


Figure 4 — Average days of stock cover by category ($\text{Physical Stock} \div \text{Average Daily Sales}$)

The inventory register confirms exactly the imbalance described in Problem Statement 1. Food carries only 12.3 days of average stock cover — on average, the store holds less than two weeks of buffer stock for its fastest, most frequently sold category. Cleaning (54.0 days), Personal Care (48.4 days), and Stationery (51.4 days) sit at roughly 4× that level. Twenty-three stockout instances (physical count = 0) were recorded across the observation window; 12 of these were Food items, against 7 in Cleaning + Personal Care (2 + 5) and 4 in Stationery. 24 products sit on more than 45 days of stock cover, led by Color Pencil, Vim Liquid, and others with over 110 days of cover. Together these slow movers hold roughly ₹23,642, in stock value, of which ₹17539.55 sits in Cleaning + Personal Care alone versus just ₹2,311 in Food.

This is a direct, quantified version of Problem Statement 1: fast-moving Food items stock out repeatedly while slow-moving Cleaning, Personal Care, and Stationery items sit overstocked, tying up cash that could instead fund more frequent Food restocking. An ABC-style split — treating Food as the store’s “A” (high-velocity) category and the 24 slow movers above 45 days of cover as “C” items — gives a direct, actionable reorder-priority list for the final report.

5.3. Margin Leakage — Problem Statement 2

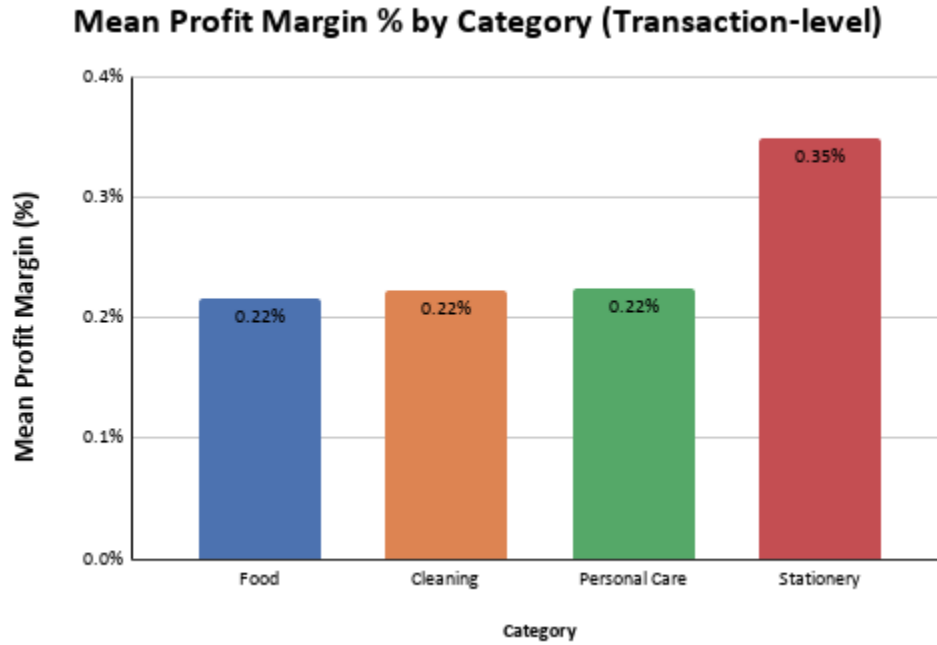


Figure 5 — Mean transaction-level profit margin (%) by category

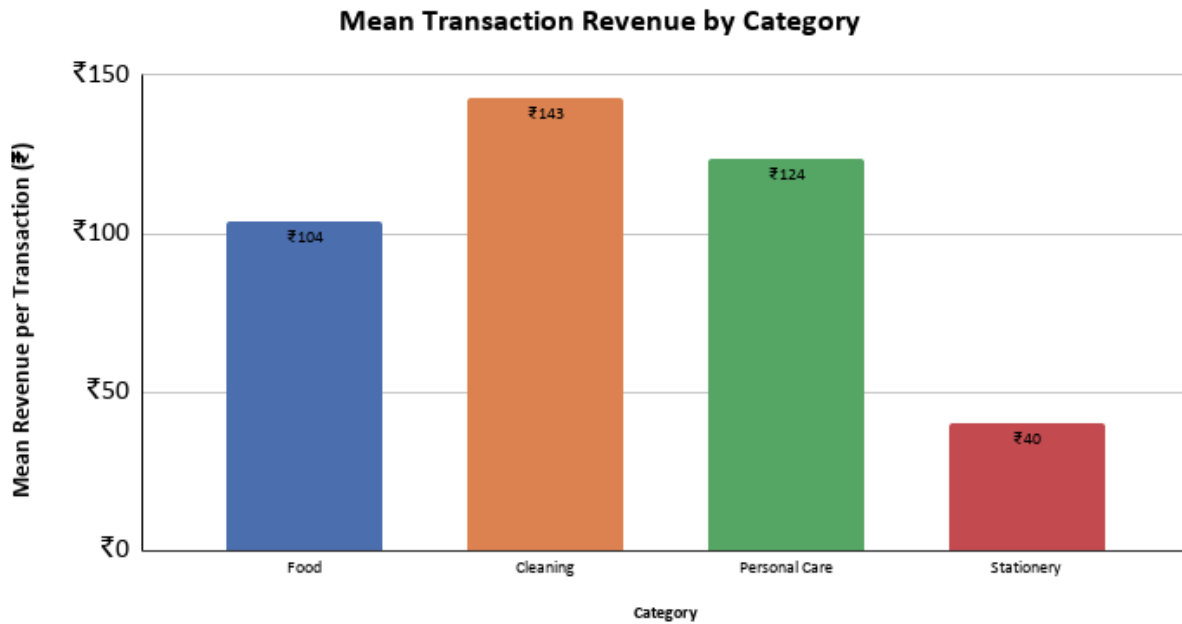


Figure 6 — Mean transaction-level revenue (₹) by category

Table 3.2.3 already shows Stationery earning a mean margin of 34.94%, roughly 13 percentage points above Food, Cleaning, and Personal Care, which cluster tightly around 21.7–22.4%. Food specifically shows the weakest position: the lowest mean margin among the four categories (21.68%) combined with the highest wastage — 29 units recorded as wasted/damaged in the Inventory register across the period, versus 8 in Personal Care, 1 in Cleaning, and 0 in Stationery, consistent with Food being the only meaningfully perishable category. The same understocking pattern identified in Section 5.2 likely costs the store additional lost sales on days Food items are unavailable (12 stockout instances) — a cost that doesn’t show up directly in the revenue figures but shows up as a forgone margin. Together, spoilage losses and stockout-driven lost sales are the most plausible drivers of Food’s margin gap, and both point to the same fix: smaller, more frequent Food restocking instead of the bulk, occasional purchase pattern implied by its low days-of-cover baseline.

5.4. Cross-Category Selling Potential — Problem Statement 3

Customer_Type	Essentials %	Food %	Stationery%	count
Student	43.1	41.2	15.6	428
Hostel Resident	53.8	40.9	5.3	283
Local Resident	69.9	25.7	4.4	158
Staff	39.1	47.2	13.7	55

Table 5.4.1 —Customer-segment category affinity (% of each segment’s total spend)

Because the Sales data does not currently have a transaction/basket ID, cross-category buying was approximated by treating all items bought by the same customer type on the same day as a proxy “basket.” Of 150 such customer-type/day groupings, 141 (94%) included purchases from more than one category. The most frequent category pairings were Food + Personal Care (66.0% of multi-category baskets), Food + Stationery (56.7%), and Personal Care + Stationery (48.2%); Cleaning paired least often with Stationery (35.5%). Table 5.4.1 adds a segment-level view: Local Residents skew heavily toward Essentials (69.94% of spend) with the least Stationery exposure, while Staff show the highest Food share (47.17%) and the most balanced spread across all three groups — useful context for deciding where to place any cross-category bundle.

How the pairing percentages were derived: for each of the 141 multi-category baskets, every pair of categories present in that basket was counted once — e.g. a basket containing Food, Personal Care, and Stationery contributes to three pairs (Food–Personal Care, Food–Stationery, Personal Care–Stationery). Each pair’s count was then divided by 141 (the total number of multi-category baskets) to get the percentage below. Table 5.4.2 shows this calculation in full so every percentage cited above is traceable back to a raw count.

Category_Pair	Co-Purchase_Count	Multi-Category B
Food + Personal Care	93	66
Food + Stationery	80	56.7
Personal Care + Stationery	68	48.2
Food + Cleaning	68	48.2
Cleaning + Personal Care	61	43.3
Cleaning + Stationery	50	35.5

Table 5.4.2 — Category-pair co-occurrence within multi-category baskets

While this proxy cannot confirm that the same customer bought both items in a single visit with certainty, the pattern is suggestive: Food and Personal Care are the store's most natural cross-sell pair, ahead of Food and Stationery. This points toward testing shelf placement or small bundling offers that put snack/beverage items near personal-care essentials, and stationery near the checkout for students. A firm answer requires adding a transaction ID which a small retail shop doesn't have .

5.5. Summary of Recommendations

- Increase restocking frequency for Food (target closer to 20–25 days of cover instead of 12) and reduce bulk order sizes for slow-moving Cleaning/Personal Care/Stationery SKUs sitting above 45 days of cover, to free up the ₹17539.55 currently tied up in overstocked Cleaning + Personal Care alone.
- Investigate spoilage drivers for the highest-wastage Food SKUs and consider markdown/clearance triggers before items approach expiry, to recover margin currently lost to write-offs.
- Review Food, Cleaning, and Personal Care pricing against the 34.94% margin Stationery already achieves — even a partial narrowing of this gap would meaningfully lift blended store margin above the current 22.05%.
- Combining Food + Personal Care shelf adjacency or small bundle offer, and track can overall increase the profit revenue.
- Overall gross profit across the store was ₹20,210 on ₹91,675 of revenue (19.2% net margin) by assumption Fixed operating costs (rent, electricity,) is Rs 2650 over three months period by having conversation with the owner .